University of Idaho
Department of Computer Science
Coeur d'Alene

CS455/555

Machine Vision: Syllabus & Course Overview

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August 19, 2024

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Faculty

John Shovic - Professor

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Garrett Wells - Lecturer/TA

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Personal Introduction

- 2022 B.S.C.S., Moscow
- 2022-23 M.S.C.S., CDA, Project Vineheart
- 2024-now Ph.D. Project Evergreen

Course Overview

Syllabus, posted on Canvas

Course Materials

Required Materials

1. Laptop
   - Python >= v3.10
   - OpenCV v4.10
2. Webcam (integrated/USB)

Course Materials

Provided Resources

Canvas!
Lecture Slides

Course Materials

Miscellaneous Resources

- OpenCV
- Python
- Anaconda
- OpenCV Documentation
- Pixabay

Grading Policy

Undergraduate Sections

Final Project	20%
Midterm & Final Exams	25%
Assignments	55%

Graduate Sections

Midterm & Final Exams	15%
Final Project	30%
Assignments	55%

Assignments

Basic Summary

- 9 Assignments
- Due every 2 weeks
- Final portion of semester, just work on final project

Final Project

Overview

- Larger project: research + proposal + work + presentation
- Proposals approved by Dr. Shovic

Final Project

Project Examples

- Automated Minesweeper
- IFG Stapler
- Plate Food Estimation
- Gesture Recognition
- Eye-Controlled Computer

Course Overview

Recap

- Lecture
- Assignments approx. every 2 weeks
- Exams
- Final Project

Course Topic Overview

1. Background: How do machines see the world?
   - Human Vision
   - Cameras
2. Image Processing Techniques
3. AI Techniques
   - Statistical inference
   - Mathematical techniques
   - ML

Conceptual Introduction to Machine Vision

What is Robot and Machine Vision?

- Humans generally rely on sight as their dominant sense.

Conceptual Introduction to Machine Vision

What is Robot and Machine Vision?

- Humans generally rely on sight as their dominant sense.
  - depth
  - color

Conceptual Introduction to Machine Vision

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- Humans generally rely on sight as their dominant sense.

- We want robots/software to utilize similar information to perceive and make decisions.

Conceptual Introduction to Machine Vision

What is Robot and Machine Vision?

- Humans generally rely on sight as their dominant sense.

- We want robots/software to utilize similar information to perceive and make decisions.
- *“Machine Vision”* or *“Computer Vision”* encapsulates the techniques used to do this.

Conceptual Introduction to Machine Vision

What is Robot and Machine Vision?

- Humans generally rely on sight as their dominant sense.

- We want robots/software to utilize similar information to perceive and make decisions.
- *"Machine Vision"* or *"Computer Vision"* encapsulates the techniques used to do this.
- Examples and applications in this class will come primarily from robotics.

Machine Vision Software

- OpenCV

Machine Vision Software

- OpenCV
  - 1999 Intel, later Willow Garage

Machine Vision Software

- OpenCV
- Spin-offs

Machine Vision Software

- OpenCV
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  - BoofCV – *faster, strongly typed OpenCV*

Machine Vision Software

- OpenCV

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  - BoofCV – *faster, strongly typed OpenCV*
  - SimpleCV – *simpler OpenCV*

Machine Vision Software

- OpenCV
- Spin-offs
  - BoofCV – *faster, strongly typed OpenCV*
  - SimpleCV – *simpler OpenCV*
  - MATLAB

Machine Vision Software

- OpenCV
- Spin-offs

- Deep Learning Specific Software

Machine Vision Software

- OpenCV
- Spin-offs
- Deep Learning Specific Software
  - CUDA
  - TensorFlow
  - Keras
  - Caffe
  - MATLAB
  - OpenVINO
  - DeepFace

Assignment #1

Overview